

FMPTCP: Achieving High Bandwidth Utilization and Low Latency in Data Center Networks

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Abstract—The utilization of Multi-path TCP (MPTCP) has been demonstrated to provide superior transport-layer support for data center networks (DCNs) due to its exceptional resource utilization and load-balancing capabilities. However, the substantial path diversity can make it challenging to utilize network resources to their full potential in DCNs. This paper focuses on studying the resource allocation issue of MPTCP from a resource optimization perspective. Based on theoretical analysis, we propose FMPTCP, which uses a feedback-based congestion control algorithm (FCC) and a feedback-based multi-path routing algorithm (FMP) to jointly achieve high bandwidth utilization and low round-trip time (RTT) in DCNs. The FCC algorithm utilizes probabilistic explicit congestion notification (ECN) to provide feedback on path congestion degree, and uses a gradient descent method to adjust the congestion window for optimal resource utilization and load balancing under a fixed routing topology. On the other hand, the FMP algorithm employs a hop-by-hop feedback mechanism to notify in-network congestion and path delay information, allowing for transparent multi-path routing for MPTCP flows. Our extensive simulations demonstrate that FMPTCP enables effective network resource utilization, which not only enhances overall throughput but also reduces transmission latency for DCNs.

Index Terms—Multi-path TCP, data center network, congestion control, multi-path routing.

I. INTRODUCTION

THE prevalence of cloud computing and storage has made data center networks (DCNs) increasingly critical in

modern networks [1]. As more businesses deploy web services and remote storage, the scale of current DCNs has expanded, with some containing tens of thousands of servers hosting distributed applications. Consequently, communication within these networks is distributed across multiple racks of machines in DCNs [2], [3]. This presents a significant challenge due to the limited network resources such as bandwidth. The large amount of traffic within data centers can place excessive pressure on DCNs [4]. Therefore, efficient transport-layer protocols are urgently needed to meet the increasing demand for various applications [5], [6].

Unlike traditional networks such as the Internet, modern large-scale DCNs have a more redundant structure and multiple routing paths between end hosts to alleviate communication overhead bottlenecks. As a result, single-path transmissions may underutilize network bandwidth while leaving many links idle. In contrast, multi-path transmissions can significantly improve bandwidth utilization by leveraging more available paths in the network [7]. Multi-path TCP (MPTCP) is one of the most popular transport-layer protocols that uses multiple paths concurrently between a pair of end hosts to improve bandwidth resource utilization. MPTCP has proven to be an efficient transport protocol in various application scenarios, especially for long flows in the network [8], [9], [10], [11]. MPTCP has great potential to improve resource utilization and provide better load balancing compared to traditional single-path transport-layer protocols, thanks to the path diversity present in DCNs [12], [13].

MPTCP has the potential to enhance performance in DCNs as compared to single path transmission protocols, but there is still significant room for improvement in terms of resource utilization optimization. The current designs of coupled congestion control algorithms for MPTCP primarily focus on end-to-end transmission performance, which makes it challenging to allocate rates while taking into account the entire network optimization. Moreover, loss-based coupled congestion control algorithms tend to experience unstable transmission rates and significant fluctuations due to packet loss's random nature. Furthermore, MPTCP can only balance congestion among its established subflows through end-to-end congestion control. In DCNs, the number of available paths between end hosts increases rapidly, while the number of subflows in an MPTCP connection is limited. It is difficult

Manuscript received 24 October 2022; revised 10 April 2023 and 9 August 2023; accepted 29 September 2023. Date of publication 11 October 2023; date of current version 17 January 2024. This work of Jiangping Han, Kaiping Xue, Jian Li, and Yitao Xing was supported in part by the National Natural Science Foundation of China (NSFC) under Grant No. 62302472, the Fundamental Research Funds for the Central Universities under Grant No. WK2100000039, Xiaomi Young Talents Program, and Youth Innovation Promotion Association of Chinese Academy of Sciences (CAS) under Grant No. Y202093. The associate editor coordinating the review of this article and approving it for publication was G. Iosifidis. (*Corresponding author: Kaiping Xue.*)

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Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TCOMM.2023.3323670>.

Digital Object Identifier 10.1109/TCOMM.2023.3323670

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for MPTCP connections to utilize all available paths with different subflows. Then, determining the paths of subflows in the network is another key for optimizing resource utilization efficiency. To address this issue, routing schemes in the network must be considered jointly with end-to-end MPTCP connection information. MPTCP, unlike TCP, encounters multiple shared bottlenecks (i.e., some subflows share the same bottlenecks) in the network. Shared bottlenecks often result in resource allocation conflicts and underutilization of network resources. The fabric topology of DCNs and the large number of subflows make shared bottlenecks in MPTCP connections more likely, which should be considered in in-network routing. Previously distributed routing algorithms only consider traffic balance within the network and do not take into account shared bottlenecks within MPTCP connections. Although some SDN-based solutions [14], [15] can take global information into consideration for enhancing path disjointness, they typically have high computational overhead and control latency. As a result, better support is needed to jointly improve the overall transmission performance of DCNs.

In addition, ensuring low latency and queue size is a critical requirement for DCNs, as queuing delay can significantly impact data transmission performance [16], [17]. Thus, any transmission optimization design for DCNs must prioritize both high bandwidth utilization efficiency and low delay for the entire network. This study focuses on analyzing the rate allocation and multipath routing problems of MPTCP, treating MPTCP transmission as a network optimization problem that considers both transmission rate and path delay. The study reveals that in-network feedback information is essential to maximize network resource utilization and that multi-path routing within the network and data allocation within MPTCP connections must be jointly considered. To address these concerns, we propose FMPTCP, a feedback-based MPTCP for DCNs that includes a congestion control algorithm (FCC) and a multi-path routing algorithm (FMP) to achieve optimal rate allocation and multi-path routing. FCC utilizes a probabilistic explicit congestion notification (ECN) [18] mechanism to signal and feedback in-network congestion levels, which are readily available in current DCN infrastructures [19]. FMP is an in-network hop-by-hop multipath routing algorithm further designed for MPTCP. By exploiting congestion and delay information in the network and jointly making routing decisions with consideration of network congestion and MPTCP shared bottlenecks, FMP further improves network resource utilization. With the cooperation of FCC and FMP, FMPTCP achieves high bandwidth utilization and low delay in DCNs. The major contributions of this paper are summarized as follows.

- The optimization problem of multipath transmission in DCNs is modeled to consider both throughput and delay. The problem is divided into rate allocation and multi-path routing sub-problems. Based on theoretical analysis, we propose FMPTCP, which includes both FCC and FMP to jointly enhance the performance of DCNs.
- FCC and FMP work together to optimize the performance of DCN. FCC uses a probabilistic ECN feedback mechanism to adjust the rate allocation within each

MPTCP connection, aiming to improve throughput and reduce transmission latency. Meanwhile, FMP routes subflows through low-congestion paths and avoids MPTCP shared bottlenecks, thereby enhancing the resource utilization efficiency of the entire network.

- We show, through simulation experiments, that FMPTCP outperforms the traditional MPTCP in terms of throughput and delay, while improving the overall resource utilization in DCNs. Compared with previous algorithms, FMPTCP increases the throughput by 12.4-35.9% and reduces the transmission latency by 32.5-51.2% in DCNs.

The rest of this paper is organized as follows. A brief review of related work is shown in Section II. Section III introduces the drawbacks and challenges of multi-path transmission in DCNs. Section IV presents the formulation and analysis of the multi-path transmission problem in DCNs. Section V presents the design of FMPTCP, including FCC and FMP. Section VI presents the simulation results. Finally, Section VII concludes the paper.

II. RELATED WORK

The design of the transport protocol in DCNs has been studied by many researchers. Traditional single-path transport protocol, such as TCP, does not perform well for both long and short flows in DCNs. Advanced single-path transport protocols [16], [17], [19], [20], [21], [22] have been developed to support the transmission requirements of high bandwidth and low delay in DCNs. DCTCP [19], D²TCP [23], and DCQCN [20] enable ECN to notify congestion in advance to improve bandwidth utilization. Timely [21], HPCC [22], Swify [24], and powerTCP [17] utilize high-precision delay detection and in-band network telemetry (INT) to control the end-to-end transmission and further achieve low delay performance.

Compared with single-path protocols, MPTCP [13] provides better network resource utilization in large-scale DCNs. Raiciu et al. [13] verified that MPTCP can improve network utilization in DCNs, and designed LIA to couple congestion control strategies running on different subflows through the growth rate of correlation subflows. Several congestion control variants emerged after LIA, including OLIA [25], which determines subflow conditions based on packet loss intervals and adds an offset to the congestion window increment. BALIA [26] optimizes MPTCP in a network using a fluid model, while CADIA [27] considers shared bottlenecks in coupled congestion control. However, these algorithms rely on packet loss for congestion window adjustment, resulting in high transmission rate fluctuations and long queue delays [10], [28]. There are also some DCN-specific MPTCP variants. To address these issues and improve throughput and delay performance in DCNs, ECN-based multi-path solutions have been proposed. XMP [29] uses an ECN-based coupled congestion window adjustment method with both RTT and congestion window on all subflows to balance congestion, ensuring equal congestion levels on each subflow. AMP [30] analyzes excessive resource usage of MPTCP in incast scenarios, detecting and keeping only one active subflow to

TABLE I
SUMMARY OF RELATED ALGORITHMS

Solution	Transport	Congestion Control	Path Management	In-network Routing	Performance Metrics (QoS)
DCTCP	TCP	Single-path, ECN-based	×	×	Throughput/Queue
D ² TCP	TCP	Single-path, ECN-based	×	×	Throughput/Queue
DCQCN	RDMA	Single-path, ECN-based	×	×	Throughput/Queue
TIMELY	RDMA	Single-path, Delay-based	×	×	Throughput/Queue
Swift	RDMA	Single-path, Delay-based	×	×	Throughput/Queue
HPCC	RDMA	Single-path, INT-based	×	×	Throughput/Queue
PowerTCP	RDMA	Single-path, Delay- & INT-based	×	×	Throughput/Queue
LIA	MPTCP	Multi-path, Loss-based	×	×	Throughput/Fairness
OLIA	MPTCP	Multi-path, Loss-based	×	×	Throughput/Fairness
BALIA	MPTCP	Multi-path, Loss-based	×	×	Throughput/Fairness
CADIA	MPTCP	Multi-path, Loss-based	×	×	Throughput/Fairness
XMP	MPTCP	Multi-path, ECN-based	×	×	Throughput/Queue/Fairness
AMP	MPTCP	Multi-path, ECN-based	✓	×	Throughput/Queue/Fairness
MMPTCP	MPTCP	Multi-path, Unspecified	✓	×	Throughput/Short flows
DCMPTCP	MPTCP	Multi-path, ECN-based	✓	×	Throughput/Short flows
ECMP	×	×	×	Random hash	Resource Utilization
CONGA	×	×	×	Path congestion-based	Resource Utilization
DRILL	×	×	×	Local queue-based	Resource Utilization
MLAB	×	×	×	Global load-based	Resource Utilization

reduce resource usage. MMPTCP [31] uses a packet scattering technique to improve delay performance of short flows while it acts as a regular MPTCP for long flows. DCMPTCP [32] includes a path management mechanism to identify rack-local traffic and eliminate unnecessary subflows, enhancing the performance of rack-local flows or many-to-one short flows.

Improving network resource utilization in DCNs also requires addressing the routing problem [33], which is another critical issue. The typical CLOS structure of DCN network topology extends multiple paths between host pairs [34], making traditional routing strategies that select only one fixed route insufficient and leading to underutilization of network paths. To take advantage of all available paths in the network, the equal-cost multi-path (ECMP) [33] routing strategy is deployed to route flows across multiple paths. ECMP is a hop-by-hop flow-based load balancing strategy that distributes flows evenly across different paths, treating each path equally and randomly selecting routing paths, which is suitable for both single-path and multi-path transmission protocols. However, ECMP does not account for in-network congestion information and is susceptible to hash conflicts and network asymmetries. CONGA [35], Drill [36], and MLAB [37] further consider routing based on the congestion information in the network. These algorithms either use a packet to record the maximum congestion on the path, or use the queue length of the local port to make routing decisions. However, most previous routing schemes are designed for single-path transport protocols and do not consider multiple subflows in the same connection, which may not achieve optimal performance. The related algorithms are summarized in Table I.

III. DRAWBACKS AND CHALLENGES OF UTILIZING MULTI-PATH TRANSMISSION IN DCNs

Due to the significant path diversity present in DCNs, MPTCP flows have multiple alternative paths to select from and migrate data, enabling better load balancing in the network. Fig. 1 illustrates a three-tier DCN with 4 core switches,

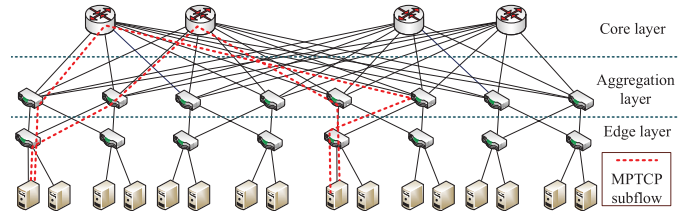


Fig. 1. MPTCP flows with multiple subflows over different paths in DCNs.

8 aggregation switches, and 8 edge switches, where up to 16 different paths can exist between a pair of end hosts. As the network size expands, the number of available paths grows exponentially. Some paths are link disjoint, while others have shared links. By exploring more paths in the network, MPTCP can distribute traffic and utilize multiple paths to improve resource utilization efficiency. Since MPTCP can select several paths to establish subflows, and balance the load to improve the transmission performance of a single connection and the resource utilization efficiency of the entire network, the multi-path routing and the rate allocation will be the key to determine the performance of DCNs. Yet, how to select paths and allocate rates to maximize the network utility is still a challenging problem that needs to be solved.

From the viewpoint of congestion control load balancing within MPTCP connections, traditional MPTCP may not always be Pareto optimal, as discussed in [25]. We have further expanded this problem analysis to consider both end-to-end congestion control and in-network routing. Selecting non-optimal routing paths and causing underutilization of shared bottleneck bandwidth may result in suboptimal performance. Therefore, achieving the best performance requires careful consideration of both end-to-end congestion control and in-network routing. As shown in Fig. 2, we use a simplified topology to illustrate that MPTCP cannot be optimal. Each link between two nodes has a bandwidth of 1 Gbps. When following the path shown in the figure, it may eventually lead to suboptimal performance. Considering max-min bandwidth allocation [38] for subflows, where the bandwidth allocated

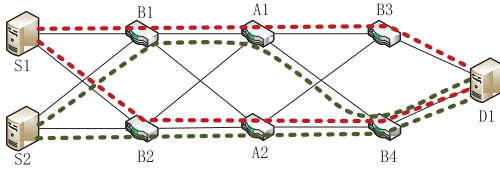


Fig. 2. An example of MPTCP in data center networks.

on each subflow is at least the same as the subflows passing through the most congested bottleneck link, each of the three subflows passing through B4-D1 gains 0.33 Gbps. Then, the subflow S1-B1-A1-B3-D1 gains 0.67 ($=1 - 0.33$) Gbps. Also, the MPTCP connection between S1 and D1 (MPTCP 1) contains two subflows that pass through S1-B1-A1-B3-D1 (subflow 1, with a bandwidth allocation of 0.67 Gbps) and S1-B2-A2-B4-D1 (subflow 2, with a bandwidth allocation of 0.33), respectively. The MPTCP connection between S2 and D1 (MPTCP 2) contains two subflows that pass through S2-B1-A2-B3-D1 (subflow 1, with a bandwidth allocation of 0.33 Gbps) and S2-B2-A2-B4-D1 (subflow 2, with a bandwidth allocation of 0.33 Gbps), respectively. Thus, MPTCP 1 achieves a throughput of 1 Gbps and MPTCP 2 achieves a throughput of 0.66 Gbps. The overall throughput of two MPTCP sessions is 1.66 Gbps, which is less than the overall available bandwidth (2 Gbps). In general, if MPTCP 2 allocates x Gbps on subflow 1, the overall throughput of two MPTCP flows will be $(2 - x)$ Gbps, which wastes x Gbps bandwidth resources.

The suboptimal outcome observed is attributed to both suboptimal rate allocation and multi-path routing. Specifically, Path 1 of S2-D1 shares the bottleneck of both B1-A1 and B4-D1, leading to resource utilization on both sides. Allocating data on subflow 1 of MPTCP 2 can result in underutilization of network resources, and therefore, MPTCP 2 should minimize data on its subflow 1 to optimize rate allocation. In addition, to ensure fairness between MPTCP 1 and 2, MPTCP 1 should minimize the data on its subflow 2, resulting in both MPTCP flows gaining 1 Gbps. Furthermore, changing the MPTCP connection between S2 and D1 to use B3 in the first path can result in both connections receiving higher bandwidth. When it comes to multi-path routing, two important factors need to be taken into consideration. Firstly, low-congested links should be given priority in the routing path to maximize the utilization of available network resources. Secondly, shared bottlenecks among subflows must be avoided to reduce underutilization of bandwidth within the same MPTCP connection. This suboptimal utilization problem is more serious in DCNs than in Internet. In DCNs, MPTCP often employs more subflows to achieve high network resource utilization efficiency [39], shared bottlenecks are more likely to occur and will significantly hinder the overall performance. This paper proposes a joint optimization approach that considers two aspects: congestion control and multi-path routing to address this issue in DCNs. The existing MPTCP solution is limited as it is an end-to-end solution that lacks a comprehensive view of the entire network, hindering overall optimization. Similarly, in-network routing is also insufficient

as it does not have a view of end-to-end transmissions, leading to the inability to avoid shared bottlenecks, which in turn reduces transmission performance. Hence, joint optimization is necessary to optimize resource utilization in DCNs.

IV. PROBLEM FORMULATION AND ANALYSIS

A. Network Model

We model the network as a graph, similar to the approach taken in [40], which assumes a static network condition over a short period of time. To account for the dynamic nature of network traffic, the network environment is divided into several intervals, where each interval is considered to have a static network model. The transition from one time slice to another is then viewed as a shift from an equilibrium state of the old model to a non-equilibrium initial state of the new model, capturing the inherent dynamics of network traffic.

The network consists of a link set $\mathcal{L} = \{L_i | i = 1, \dots, n\}$, routing-path set $\mathcal{R} = \{R_j | j = 1, \dots, s\}$, and MPTCP flow set $\mathcal{S} = \{S_k | k = 1, \dots, r\}$. Link L_i refers to a point-to-point physical link. R_j refers to an end-to-end routing path, comprising of multiple links and nodes connected in a chain. S_k is a path set of MPTCP, which comprises of multiple individual paths. Each link L_i has finite capacity $c_i > 0$ and delay $d_i > 0$. We use $\mathbf{c} = \{c_i\}$ to denote the collection of the capacity of each link. The relationship between $\mathcal{L}$ and $\mathcal{R}$ is represented by a matrix $\mathbf{A} = \{a_{ij}\}$, where $a_{ij} = 1$ if $L_i \in R_j$, and $a_{ij} = 0$ otherwise. Also, the relationship between $\mathcal{R}$ and $\mathcal{S}$ is represented by a routing matrix $\mathbf{B} = \{b_{jk}\}$, where $b_{jk} = 1$ if $R_j \in S_k$, and $b_{jk} = 0$ otherwise. $\mathbf{B}$ can be represented by two matrices $\mathbf{B} = \mathbf{F} * \mathbf{G}$, where $\mathbf{F} * \mathbf{G}$ is the hadamard product and $\mathbf{B} = \{b_{jk} | b_{jk} = f_{jk} * g_{jk}\}$. $\mathbf{G} = \{g_{jk}\}$ is the relationship between paths and available paths of MPTCP flows, where $g_{jk} = 1$ means R_j is an available path that S_k can choose, and $g_{jk} = 0$ otherwise. $\mathbf{F} = \{f_{jk}\}$ is the multi-path routing matrix, which denotes the relationship between available paths and selected paths of MPTCP flows, where $f_{jk} = 1$ means path j is selected, and $f_{jk} = 0$ otherwise.

Let $\mathbf{x} = \{x_{jk}\}$ denote the rate matrix, where $x_{jk} \geq 0$ is the rate of path R_j of MPTCP flow S_k , and $U_k(\cdot)$ denote the utility function of S_k , which is strictly concave and twice continuously differentiable in the nonnegative domain, and $U_k(0) = -\infty$. A widely used utility for α -fairness is $U_\alpha(y) = -\frac{y^{(1-\alpha)}}{1-\alpha}$, where $U_1(y) = \log(y)$ achieves proportional fairness [41]. In general, the utility function can be defined as a function that relates to the total bandwidth of all its paths, which means $U_k(\sum_j b_{jk} x_{jk})$. The performance of MPTCP is significantly influenced by path delay, which in turn can be significantly affected by asymmetric delays. Hence, when modeling MPTCP, it is crucial to consider the impact of asymmetric path delay.

We add the time delay asymmetry ratio $\frac{\ell_{min}^k}{\ell_j^k}$ as a weight of each subflow, where $\ell_j = \sum_i a_{ij} d_i$ is path delay of R_j , and $\ell_{min}^k = \min_{R_j \in S_k} \ell_j$ is the minimum path delay of S_k .

The objective of the MPTCP problem is to determine a multi-path routing matrix $\mathbf{F}$ and a rate matrix $\mathbf{x}$ so as

to maximize the total utility subject to capacity constraints. Thus, we formulate the problem as:

$$\min_{\mathbf{x} \geq 0, \mathbf{F}} - \sum_{k=1}^r U_k \left(\sum_{j=1}^s b_{jk} \frac{\ell_{\min}^k}{\ell_j} x_{jk} \right) \quad (1)$$

$$s.t. \quad \mathbf{AB}^T \mathbf{x} \leq \mathbf{c}, \quad (1a)$$

$$\mathbf{B} = \mathbf{F} * \mathbf{G}, \quad (1b)$$

$$\|\mathbf{f}_k\|_0 \leq n_k, \quad k = 1, \dots, r, \quad (1c)$$

where inequality $\mathbf{AB}^T \mathbf{x} \leq \mathbf{c}$ means that each entry in $\mathbf{AB}^T \mathbf{x}$ is less than the corresponding entry in $\mathbf{c}$. Constraint (1a) means that the total traffic carried on each link cannot exceed the link capacity. $\|\cdot\|_0$ is the 0-norm, which represents the number of non-zero elements in the vector. Constraint (1c) means MPTCP flow k is able to select n_k paths to establish n_k subflows, where $\mathbf{f}_k$ is the k -th column of $\mathbf{F}$.

B. Problem Decomposition

Our proposed theoretical analysis aims to identify the flows that can maximize network performance. In TCP/IP networks, routing solutions dictate the physical paths for dissemination. Within each flow, transport solutions such as MPTCP determine the transmission rate in the individual paths via a congestion control algorithm. To simplify the analysis, we break down the original optimization problem into two sub-problems: *the rate allocation problem* and *the multi-path routing problem*.

1) *Rate Allocation Problem*: Decide a rate matrix $\mathbf{x}$ with a fixed routing matrix $\mathbf{B}$ to optimize the network utility.

$$\min_{\mathbf{x} \geq 0} - \sum_{k=1}^r U_k \left(\sum_{j=1}^s b_{jk} \frac{\ell_{\min}^k}{\ell_j} x_{jk} \right) \quad (2)$$

$$s.t. \quad \mathbf{AB}^T \mathbf{x} \leq \mathbf{c}.$$

The optimization function is a convex function, and the feasible region is a convex set, so it is a convex optimization problem. Let $y_k = \sum_j b_{jk} \frac{\ell_{\min}^k}{\ell_j} x_{jk}$, then the Lagrangian function is:

$$\begin{aligned} L(\mathbf{x}, \lambda) &= - \sum_{k=1}^r U_k(y_k) + \sum_{i=1}^n \lambda_i \left(\sum_{j=1}^s a_{ij} \sum_{k=1}^r b_{jk} x_{jk} - c_i \right) \\ &= \sum_{k=1}^r \left\{ -U_k(y_k) + \left(\sum_{i=1}^n \lambda_i a_{ij} \right) \sum_{j=1}^s b_{jk} x_{jk} \right\} \\ &\quad - \sum_{i=1}^n \lambda_i c_i, \end{aligned} \quad (3)$$

where $\lambda = \{\lambda_1, \dots, \lambda_n\}$ is the Lagrangian multiplier. Let $q_j = \sum_{i=1}^n \lambda_i a_{ij}$, the dual problem of the original problem can be written as:

$$\max_{\lambda \geq 0} D(\lambda) = \sum_{k=1}^r D_k(\lambda) - \sum_{i=1}^n \lambda_i c_i, \quad (4)$$

where $D(\lambda)$ is divided by a set of sub-problems:

$$D_k(\lambda) = \min_{x_{jk} \geq 0} \left\{ -U_k(y_k) + \sum_{j=1}^s q_j b_{jk} x_{jk} \right\}. \quad (5)$$

Each sub-problem corresponds to a local optimality related to a flow k .

Proposition 1: The solution of $D_k(\lambda)$ satisfies:

$$\begin{aligned} U'_k(y_k) &= \min_{j \in \mathbb{J}_k} \frac{\ell_j}{\ell_{\min}^k} q_j, \\ x_{jk} &= 0, \quad p_j > p_{\min}, \\ x_{jk} &\geq 0, \quad p_j = p_{\min}, \end{aligned} \quad (6)$$

where $U'_k(\cdot)$ is the derivative of $U_k(\cdot)$, $\mathbb{J}_k$ is the set of path of flow k , $p_j = \ell_j \cdot q_j$ denotes the path price of R_j , and $p_{\min} = \min_{j \in \mathbb{J}_k} p_j$.

Proof: See Appendix A. $\square$

The dual problem (4) can be solved using *gradient projection algorithm* [42]:

$$\lambda_i(t+1) = \left[\lambda_i(t) + \gamma \left(\sum_{j=1}^s \sum_{k=1}^r a_{ij} b_{jk} x_{jk}(t) - c_i \right) \right]^+, \quad (7)$$

where $[\lambda]^+ = \max(0, \lambda)$, γ is the step size, and

$$x_{jk}(t) = x_{jk}^*(\lambda(t)), \quad (8)$$

where $x_{jk}^*(\lambda(t))$ is the solution of Eq. (5). And by duality theory, there exists a dual optimal price λ^* such that $\mathbf{x}^*(\lambda^*)$ is indeed primal optimal [7].

Eq. (8) and Eq. (7) jointly express a distributed iterative solution. Links in the network adjust the value of λ , and broadcast the value to all flows. Then, each flow updates its rate x_{jk} based on the value of λ it receives. After several iterations, the optimal point is finally reached. According to Eq. (8), each flow does not need to know the information about all the links in the network, only the feedback on its routing path is needed.

To perceive in-network information and adjust MPTCP rates, a precise feedback mechanism is needed. In Eq. (7), $(\sum_{j=1}^s \sum_{k=1}^r a_{ij} b_{jk} x_{jk}(t) - c_i)$ represents the portion of traffic that exceeds the link capacity. This portion of traffic is stored at switch and occupies the queue of a switch. If $\sum_{j=1}^s \sum_{k=1}^r a_{ij} b_{jk} x_{jk}(t) > c_i$, queue occupancy grows linearly with $a_{ij} b_{jk} x_{jk}(t) - c_i$ and vice versa. Therefore, $\lambda_i(t)$ is in directly proportion to the queue occupancy on the switch, which reflects the link congestion degree. When the queue is empty, $\lambda_i(t) = 0$. Therefore, q_j is in proportion to the cumulative queue occupancy along the routing path, which denotes the congestion degree of a path. We propose FCC to achieve distributed rate allocation in practice, where switches give the feedback information of queue occupancy along the routing path through ECN mechanism, and MPTCP senders adjust congestion window based on ECN feedback to change its rate. The algorithm in more detail can be found in the next section.

2) *Multi-Path Routing Problem*: Decide a multi-path routing matrix $\mathbf{F}$ to optimize the network utility.

$$\begin{aligned} \min_{\mathbf{F}} U(\mathbf{F}) &= - \sum_{k=1}^r U_k(y_k) \\ \text{s.t.} \quad & (1b), (1c), \\ & \mathbf{x} = \mathbf{x}^*(\mathbf{F}), \end{aligned} \quad (9)$$

where $\mathbf{x}^*(\mathbf{F})$ is the optimal solution of rate allocation problem.

Multi-path routing problem is NP-hard. This can be shown by a simple reduction from the single-path routing problem, which has been proven to be NP-hard [43]. We show that Eq. (10) is performed as an equivalence to the original multi-path routing problem:

$$\begin{aligned} \min_{\mathbf{x} \geq 0} U(\mathbf{x}) &= - \sum_{k=1}^r U_k(y_k) \\ \text{s.t.} \quad & \mathbf{A}\mathbf{G}^T \mathbf{x} \leq \mathbf{c}, \\ & \|\mathbf{x}_k\|_0 \leq n_k, \quad \forall k, \end{aligned} \quad (10)$$

where $\mathbf{x}_k$ is the k -th column of $\mathbf{x}$. If $\mathbf{x}$ is a feasible solution of (10), a multi-path routing matrix $\mathbf{F}(\mathbf{x})$ is formulated as:

$$f_{jk} = \begin{cases} 1 & x_{jk} > 0, \\ 0 & \text{otherwise.} \end{cases} \quad (11)$$

Then $\mathbf{F}(\mathbf{x})$ is also a feasible solution of (9).

Proposition 2: Let $\mathbf{x}_{opt}$ be the optimal solution of (10), $\mathbf{F}(\mathbf{x}_{opt})$ is also the optimal solution of (9).

Proof: See Appendix B. $\square$

By dropping the last nonconvex constraint in Eq. (10), this yields the rate allocation problem, and can also be viewed as a relaxation of Eq. (10). An approximate solution of Eq. (10) can be given by choosing the paths with the n_k largest rates for flow k [44]. According to **Proposition 1**, a flow allocates a positive rate on path j only if $p_j = p_{min}$. That is, the flow needs to choose paths with the smallest price to send data. Therefore, we propose a heuristic approach for the multi-path routing problem. Whenever a flow k joins the network, it selects paths as:

$$f_{jk} = \begin{cases} 1 & \text{if } p_j^{k-1} \text{ is the } n_k \text{ smallest path prices,} \\ 0 & \text{otherwise,} \end{cases}$$

where $p_j^{k-1} = \ell_j \sum_{i=1}^n a_{ij} \lambda^{*(k-1)}$, and $\lambda^{*(k-1)}$ is the optimal λ based on the previous $(k-1)$ flows. Each flow sorts path prices p_j of its available paths in ascending order and then selects the first n_k paths. Considering that flows join the network one by one and the time interval between two flows is usually long enough for congestion control to stabilize, the network can give feedback information of path prices for routing. However, this gives a cross-layer deployment that requires a combination of switch-side feedback and sender-side routing path selection, which results in additional modifications to the MPTCP protocol stack. We propose a multi-path routing algorithm running on switches that is transparent to the MPTCP layer, where the intermediate switches give feedback and select path hop-by-hop. The detail of our multi-path routing algorithm is described in Section V-B.

V. FEEDBACK-BASED CONGESTION CONTROL AND MULTI-PATH ROUTING ALGORITHMS

Based on the above analysis, we design FMPTCP, which consists of *feedback-based congestion control* (FCC) and *feedback-based multi-path routing* (FMP). Through the joint efforts of FCC and FMP, FMPTCP was able to achieve exceptional resource utilization and low delay in DCNs.

A. Feedback-Based Congestion Control

FCC utilizes explicit congestion notification (ECN) to provide precise and real-time feedback without excessive modification to the existing MPTCP protocol. The characteristics of DCNs allow administrators to uniformly configure ECN with the same parameters on all switches. Switches mark packets with ECN marks to reflect the congestion degree in the network. Following this, the MPTCP sender adjusts its congestion window based on the ratio of ECN marks in ACKs to manage the transmission rate and maintain a balance in congestion.

1) *ECN Feedback*: In order to maximize network utility, the source of MPTCP needs to sense the congestion of the path to adjust its rate. ECN is a common protocol for switches and routers in the network to explicitly indicate in-network congestion to source nodes. ECN is currently supported by the majority of routers and switches. The switch marks the arriving data packets with the ECN to indicate in-network congestion. The MPTCP/TCP receiver feeds its received ECN back to the sender through ACK packets.

The ECN setting on switches utilizes two threshold, *low_thresh* and *high_thresh*. When the queue length is in the range between two thresholds, packets are marked with ECN with a linearly increasing probability from 0 to *max_p*. FCC sets the ECN parameters customized, where *low_thresh* = 0, $0 < \text{high_thresh} < 1$, and *max_p* = *high_thresh*.¹ That is, when the queue occupancy is less than *high_thresh*, the ECN mark probability is as the same proportion as the queue occupancy rate. Otherwise, if the queue occupancy is larger than *high_thresh*, it marks all data packets with ECN to show that the link is heavily congested. Therefore, the ECN mark probability is in direct proportion to the link congestion degree λ_i and reflects the congestion degree of a link. Let ρ_i denotes the ECN mark probability of link L_i , and β_j denotes the ECN mark probability of path R_j , $\beta_j = 1 - \prod_i (1 - a_{ij} \rho_i) \approx \sum_i a_{ij} \rho_i$. If $\rho_i \ll 1$, the path congestion degree $q_j = \sum_i a_{ij} \lambda_i$ is approximately in direct proportion with the path's ECN mark probability β_j , where β_j also reflects the cumulative queue occupation rate of the path. The MPTCP sender gauges the level of congestion on each path by computing the ratio of ECN-labeled acknowledgments within a subflow, and then adjusts the transmission rate accordingly.

2) *Congestion Window Adjustment*: MPTCP senders maintain an estimate of the ECN-mark probability of each path j , denoted as β_j , which is calculated as the fraction

¹ECN can also be used with some active queue management (AQM) algorithms. The *low_thresh* and *high_thresh* can be set to queue line bytes or queue occupancy rate. We use queue occupancy rate in this paper.

of ECN-marked ACKs on subflow j in the last congestion window. β_j is updated once for every RTT as follows:

$$\beta_j \leftarrow (1 - \eta)\beta_j + \eta F, \quad (12)$$

where F is the fraction of packets that were marked in the last window, and $0 < \eta < 1$ is the weight given to new samples. We use $\eta = 1/16$ in FCC, which is identical to the parameter setting in [19]. Note that it needs several ACKs to calculate the ECN mark proportion. When the congestion window is small, it will lead to a high variance of statistics. FCC sets $w_{min} = 10$ as the minimum ECN calculation window, and calculates ECN-marked ACKs in $\max(w_{min}, w_j)$. Let w_j denote the congestion window of subflow j , n_{ECN} denote the number of ECN-marked ACKs in the last $\max\{w_{min}, w_j\}$ ACKs, FCC calculates F as:

$$F = n_{ECN} / \max\{w_{min}, w_j\}, \quad (13)$$

We show the congestion window adjustment of each subflow in a single MPTCP flow and omit the subscript k of MPTCP flow for the sake of simplicity. When a new ACK is received, the MPTCP sender adjusts its congestion window as:

$$w_j \leftarrow \max\{1.0, w_j + \frac{(\alpha_j - \beta_j)r_{tt_j}^2 \bar{w}}{r_{tt_{min}}^2 w_j}\}, \quad (14)$$

where r_{tt_j} is the round trip time (RTT) of subflow j , $\bar{w} = \sum_r w_r / n_{subflow}$ is the average congestion window, and $n_{subflow}$ is the number of subflows. Also,

$$\alpha_j = \alpha_{total} \frac{1/r_{tt_j}}{\sum_r w_r / r_{tt_r}^2}, \quad (15)$$

where $\alpha_{total} > 0$ is a key parameter which will be discussed later. Next, we will show how FCC achieves the same rate adjustment as Eq. (8).

Following existing work, we use the utility function $U_k(\cdot) = \log(\cdot)$ to achieve proportional fairness [41]. In practice, with the feedback information of λ in the network, each MPTCP flow can use the gradient projection algorithm to achieve $x^*(\lambda)$:

$$\begin{aligned} x_j(t+1) &= \left[x_j(t) + \theta \left(\frac{\ell_{min}}{\ell_j} U'(y(t)) - q_j \right) \right]^+ \\ &= \left[x_j(t) + \theta \left(\frac{\ell_{min}}{\ell_j} \frac{1}{y(t)} - q_j \right) \right]^+, \end{aligned} \quad (16)$$

where θ is the step size. Let $w_j(t)$ denote the congestion window of subflow j at time step t , and the RTT of subflow j is assumed to remain unchanged during the transmission, which is denoted as r_{tt_j} . Therefore, transmission rate of subflow j is $x_j(t) = w_j(t)/r_{tt_j}$. Let Δ_j denote the change of subflow j 's congestion window once a new ACK is received. Considering $r_{tt_{min}}$ as a time step, after one time step, w_j increases as:

$$w_j(t+1) - w_j(t) = \Delta_j \frac{w_j(t)r_{tt_{min}}}{r_{tt_j}}. \quad (17)$$

Then, the transmission rate of subflow j is adjusted as:

$$\begin{aligned} \frac{w_j(t+1)}{r_{tt_j}} &= \left[\frac{w_j(t)}{r_{tt_j}} + \frac{\bar{w}}{r_{tt_{min}}^2} \left(\frac{\alpha_{total}/r_{tt_j}}{\sum_r w_r(t)/r_{tt_r}^2} - \beta_j \right) \right]^+ \\ &= \left[\frac{w_j(t)}{r_{tt_j}} + \frac{\bar{w}\alpha_{total}}{r_{tt_{min}}^2} \left(\frac{r_{tt_{min}}}{r_{tt_j}} \frac{1}{y_k(t)} - \frac{\beta_j}{\alpha_{total}} \right) \right]^+, \end{aligned} \quad (18)$$

where $r_{tt_{min}} = \min_j r_{tt_j}$ is the minimum RTT of all subflows and $r_{tt_{min}}/r_{tt_j}$ approximately equals to ℓ_{min}^k/ℓ_j . Let $\theta = \bar{w}\alpha_{total}/r_{tt_{min}}$ and $q_j = \beta_j/\alpha_{total}$, Eq. (14) achieves the distributed rate adjustment at MPTCP sender and will converge at $x^*(\lambda)$.

The gradient projection method enforces the condition that x_j and w_j remain non-negative throughout the process. However, to maintain the activity of subflows, a minimum congestion window must be established for each subflow during MPTCP transmission. FCC sets $w_j = \max(1.0, w_j)$ to ensure that each subflow carries at least one packet. When $\beta_j < \alpha_j$, the window size is increased and vice versa. At the equilibrium, $\beta_j = \alpha_j = \frac{\alpha_{total}/r_{tt_j}}{\sum_r w_r/r_{tt_r}^2}$, which also represents the queue occupancy rate in the network. We set $\alpha_{total} = 1/baseRTT$, where $baseRTT$ is the base RTT without queuing delay in the network. When the network is highly congested and all the subflows have the minimum congestion window (that is, $w_j = 1$), we have $\beta_j \approx \alpha_{total} \frac{1/baseRTT}{\sum_r 1/baseRTT^2} \leq 1$. Furthermore, it can reduce the queue occupancy to $1/BDP$ when there is only one subflow running on the path, where BDP is the Bandwidth Delay Product (BDP). Considering a DCN with a link bandwidth of 1 Gbps and base RTT of 400 μ s,² the BDP equals 36 packets. If there is only one subflow running on the path, we have $w_j \approx BDP$ and $\beta_j < \alpha_{total} \frac{1/baseRTT}{BDP/baseRTT^2} = 0.03$. Therefore, FCC can keep a low queue occupancy in the network.

Eq. (14) is only used in the congestion avoidance phase. In the slow-start phase, FCC increases w_j by one if receiving an ACK, which is the same as what the original MPTCP does, since it requires a rapidly exponentially growing transmission rate to detect the available bandwidth of the network. Different from the original MPTCP, FCC exits the slow-start phase when $\alpha_j > \beta_j$, and sets $ss_j = \max\{1.0, w_j/2\}$, where ss_j is the slow start threshold. Moreover, when packet loss occurs or $\beta_j \geq 1$, indicating that the network is highly congested, FCC halves the congestion window once per RTT. FCC algorithm is shown in Algorithm 1. Since this corresponds to the gradient projection method, it converges to the optimal result.

B. Feedback-Based Multi-Path Routing

In this section, we propose a distributed multi-path routing algorithm, namely FMP, in which a hop-by-hop feedback and routing approach is put forward. FMP is implemented at switches and provides transparent routing for MPTCP. FCC accomplishes the optimization of both individual MPTCP

²A base RTT of 400 μ s corresponds to a link delay of 33 μ s in a 3-tier topology.

Algorithm 1 FCC Algorithm

```

if receive a new ACK on subflow  $j$  then
  Update  $\beta_j$ ;
   $\alpha_j = \alpha_{total} \sum_r \frac{1/r_{ttj}}{w_r/r_{tt_r}^2}$ ;
  if  $w_j > ss_j$  then
    /* congestion avoidance */;
     $\Delta_j = \frac{(\alpha_j - \beta_j) r_{tt_j}^2 \bar{w}}{r_{tt_{min}}^2 w_j}$ ;
     $w_j \leftarrow \max\{1.0, w_j + \Delta_j\}$ ;
  else
    /* slow start */;
     $w_j \leftarrow w_j + 1$ ;
    if  $\alpha_j < \beta_j$  then
      /* exit slow start */;
       $ss_j = \max\{1.0, w_j/2\}$ ;
if packet loss or  $\beta_j \geq 1$  on subflow  $j$  then
  halve  $w_j$  once per RTT;

```

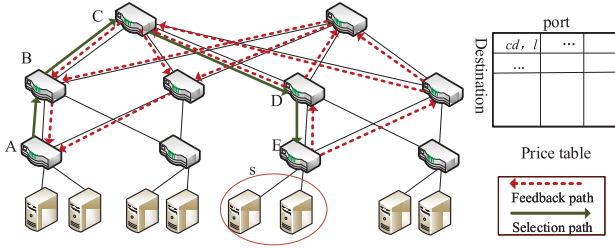


Fig. 3. Multi-path routing procession.

connections and the entire network by identifying MPTCP subflows at switches and decoupling multi-path routing from end-to-end connections. The feedback and routing processes are shown as follows.

1) *Hop-by-Hop Feedback*: FMP adopts a hop-by-hop feedback approach for perceiving the in-network congestion information. The intermediate switches periodically give feedback of the congestion information in the network and select routing path according to the feedback information. In order to save storage and computing resources, a switch will aggregate the downstream information. That is, each switch only gives feedback of the minimum price of the corresponding alternative paths to the upstream one. Finally, the intermediate switches maintain a set of path prices for each pair of destination and port for routing decision.

As shown in Fig. 3, each switch maintains a price table, and the size of which is $M \times K$, where M is the number of destinations and K is the number of switch ports. Each item in the price table contains the cumulative congestion degree and the delay of the path from the port to the destination, denoted as (cd, l) . Let s denote a destination and i denote a port of the switch, an item of price table is denoted as $(cd_i^{[s]}, l_i^{[s]})$, where $cd_i^{[s]}$ denotes the congestion degree, defined as cumulative queue occupancy rate, and $l_i^{[s]}$ is the path delay, defined as cumulative delay, along the best path from port i to destination s . Among them, the destination servers connected

to an edge switch can be aggregated into one item in the price table. That is, each s represents a set of servers connected to an edge switch. For a FatTree topology with K ports at each switch, the size of price table is $K^2/2 \times K$, which is related to $K^2/2$ edge switches and K ports.

As shown in Fig. 3, the red dotted lines represent the feedback path. A hop-by-hop feedback approach is utilized to update price tables within the network. Each switch periodically gives feedback information of congestion degree and delay to its neighbors. To be noted, a switch stores the information of each port but only gives feedback of the best port to reduce communication and storage overhead. For a destination s , the pair of congestion degree and delay of the best path from the current switch to the destination is $(cd^{[s]}, l^{[s]})$, where:

$$cd^{[s]} = cd_{i^*}^{[s]}, \quad l^{[s]} = l_{i^*}^{[s]},$$

$$i^* = \arg \min_{i \in \mathcal{D}_s} \{cd_i^{[s]} \cdot l_i^{[s]}\}, \quad (19)$$

where $\mathcal{D}_s$ is the alternative forward ports of s .

The switch periodically sends feedback information $(cd_{i^*}^{[s]}, l_{i^*}^{[s]})$ to the upstream switch. To reduce communication overhead, the switch can aggregate the feedback information for each port and send them in one feedback packet. Note that the edge switch does not need feedback information from the destination servers connected to it because there is only one path to select. In addition, it only gives $(cd_{s_0}^{[s]} = 0, l_{s_0}^{[s]} = 0)$ to the aggregation switches, where s_0 is the destination that connects to it.

When a switch receives a feedback packet, it updates the price table in its local database for future multi-path routing. Let $(cd_i^{[s]-}, l_i^{[s]-})$ denote the feedback information of destination s received from port i , the switch updates its price table as:

$$cd_i^{[s]} = cd_i^{[s]-} + Q_i,$$

$$l_i^{[s]} = l_i^{[s]-} + d_i, \quad (20)$$

where Q_i is the queue occupancy rate of port i , which reflects the congestion degree of the corresponding link, and d_i is the link delay, which can be measured by carrying a time stamp in the feedback packet.

After the feedback information is aggregated at switches, they obtain congestion degree and delay information of the best path corresponding to each port. When it needs to establish n subflows of an MPTCP connection, the multi-path routing process is hop-by-hop and started from the MPTCP sender.

2) *Multi-Path Routing*: FMP utilizes a hop-by-hop routing method implemented at switches to enable multi-path routing for MPTCP flows. In FMP, per-flow routing is accomplished by having each switch maintain an $N \times 1$ routing table per flow, where N represents the number of active subflows currently present on the switch. The per-flow routing table records the packet forwarding port for each subflow, where each subflow is identified by $\langle \text{destination address (DA)}, \text{port number (PN)}, \text{MPTCP token (MT)} \rangle$, where MPTCP “token” is a locally unique identifier given to a multi-path connection

Algorithm 2 FMP Algorithm**Algorithm:** Feedback/* Update $\{(cd_i^{[s]}, l_i^{[s]})\}$ at switch */;**for** each destination $s \in \mathcal{S}$ **do** **for** each s, i **do** **if** receive $(cd_i^{[s]-}, l_i^{[s]-})$ **then** $cd_i^{[s]} = cd_i^{[s]-} + Q_i$; $l_i^{[s]} = l_i^{[s]-} + d_i$;

/* Periodically send feedback information at each time interval */;

for each s **do** $i^* = \arg \min_{i \in \mathcal{D}_s} \{cd_i^{[s]} \cdot l_i^{[s]}\}$; $cd^{[s]} = cd_{i^*}^{[s]}, l^{[s]} = l_{i^*}^{[s]}$; Send $(cd^{[s]}, l^{[s]})$ to the upstream switches along the alternative path of destination s ;**Algorithm:** Multi-path routing**Input:** Destination: s , number of subflows: n **Output:** Selected ports for subflowsSort ports $i \in \mathcal{D}_s$ from small to large by $cd_i^{[s]} \cdot l_i^{[s]}$: $\{i_1, i_2 \dots\}$;**for** each subflow $k \in [1, n]$ **do** $j = ((k - 1) \bmod |\mathcal{D}_s|) + 1$; Forward subflow k 's packets from port i_j ;

by a host [45]. When MPTCP utilizes the “ndiffports” path manager, it establishes n subflows (where n is a parameter that can be set by the administrators) between the same pair of source and destination addresses with the same token and different port numbers. To differentiate between MPTCP flows, switches can use both the destination address and token, while port numbers can be employed to differentiate between individual subflows. When a switch receives a packet with new $\langle DA, PN, MT \rangle$, it adds a new entry to the per-flow routing table according to the routing algorithm and forwards the subsequent packets of the same subflow from the selected port to the downstream switch. If a subflow does not send data for a period of time, the routing entry is deleted from the table.

Specially, in the FatTree topology, the server has only a single IP address and cannot use multiple IP addresses to establish different subflows. MPTCP uses the same IP address and different port numbers to identify different subflows, and uses the MPTCP token to identify an MPTCP connection. The switch uses an MPTCP token to determine whether the subflow belongs to the same MPTCP connection, and the port numbers are used to distinguish between different subflows of the same connection. If the switch detects that there are several subflows with the same token, these subflows are regarded as being a part of a single MPTCP connection. For n subflows belonging to an MPTCP connection, the switch picks n ports with the n largest $cd_i^{[s]} \cdot l_i^{[s]}$ values for subflows. If the number of subflows is greater than the number of ports, the routing selection is made by round-robin. Algorithm 2 shows the FMP algorithm,

which is divided into two phases: feedback and multi-path routing.

C. Deploying FMPTCP in DCNs

This section delves into the implementation complexity and modifications required for deploying FMPTCP in DCNs. FCC and FMP are the two algorithms that comprise FMPTCP, and each necessitates distinct modifications that are supported in DCNs.

Out of the two algorithms, FCC necessitates adjustments to the ECN-based congestion window control in the MPTCP protocol stack on end nodes, as well as the implementation of a unified ECN deployment on switches. ECN is a widely supported protocol in modern switches used in DCNs, and it is already integrated into the existing TCP protocol (DCTCP in Linux kernel). FCC can leverage the existing support for ECN and update its congestion window adjustment function to modify the MPTCP stack accordingly. Deploying ECN on switches is a straightforward task as the ECN parameters can be uniformly set across all switches. The ECN mark probability is proportional to the queue length on the switches, and the computational complexity is $O(1)$. Moreover, FCC does not require any additional storage overhead on the network switches. Therefore, this implementation of FCC is feasible in a real DCN.

FMP is an in-network routing algorithm designed for DCNs, which only requires modifications to switches for hop-by-hop feedback and routing. FMP operates independently of MPTCP and does not require any modifications to the MPTCP stack. This design allows FCC and FMP to be decoupled, meaning that FCC can operate independently in cases where switches do not support FMP. FCC requires per-flow switch state and periodic hop-by-hop price feedback. The implementation can be supported by programmable switches with P4 support [46]. Additional storage and communication complexity is required for performance enhancement, and we will discuss this in the following section. Each switch maintains a price table, and the size of which is $M \times K$, where M is the number of destinations and K is the number of switch ports. For example, for a widely used FatTree topology with K ports at each switch, the size of price table is $K^2/2 \times K$. The periodic price feedback is transmitted only one hop, so the communication overhead is minimal. In a practical DCN, the storage and communication complexity overhead is acceptable for achieving further performance improvements.

Furthermore, even in the case of switch failures, FMPTCP can still ensure data transmission by utilizing multiple paths. However, the routing decisions made by FMP may become inefficient, negatively impacting the network's overall performance. Fortunately, this issue can be resolved with an easy fix. FMP can include an active path detection process based on the periodic price table feedback. Whenever a switch detects a port or link failure, it removes the corresponding entry from its price table and updates the related routing table entries. If a switch discovers that a destination or destinations are unreachable due to a port or link failure, it promptly sends a “path failure notification” to the upstream switches. Once the

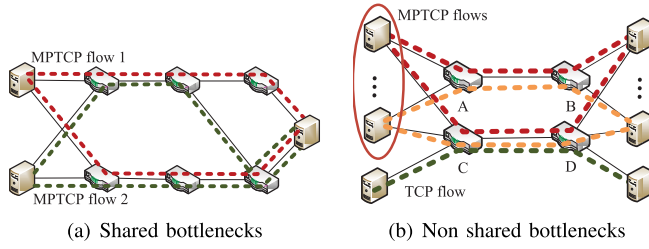


Fig. 4. Two different small network scenarios setup.

upstream switches receive the notification, they remove the affected destinations and ports from the price table and adjust the routing table entries accordingly. Lastly, if a switch does not receive the price table feedback message within a predetermined time interval (which is usually slightly longer than the feedback period to prevent misjudgment), the downstream switch is assumed to have failed. The switch then removes the corresponding port from its price table and updates the routing table entries accordingly, thereby protecting the effectiveness of routing.

VI. PERFORMANCE EVALUATION

The aim of this section is to assess the performance of FMPTCP and demonstrate the benefits it offers to DCNs, in terms of small and large-scale network environments.

A. Congestion Control in Small Scale Networks

We use ns-3 simulator [47] to evaluate the performance of different congestion control algorithms in small-scale networks as shown in Fig. 4. FCC is implemented and compared with XMP, DCTCP, BALIA, OLIA, and LIA to show its superiority. Among them, XMP, BALIA, OLIA, and LIA are coupled multi-path congestion control algorithms. DCTCP is a single-path congestion control algorithm, we run it independently on each subflow in the multi-path scenario. In the simulation, the bandwidth of each link is set to 100 Gbps, and the link delay is set to $0.5\text{--}1\text{ }\mu\text{s}$. We evaluate the performance of throughput, RTT, and queue occupancy, respectively. The parameters used in FCC are set to $high_thresh = 0.8$, $\alpha_{total} = 200$, $w_{min} = 10$, and the maximum queue length of each port of switches is set to 100 packets.³

Fig. 4(a) is a non-optimal topology with shared bottlenecks, where three subflows from different MPTCP flows go through the same bottleneck link. The bandwidth of each link is set to 100 Gbps, and the link delay is set to $0.5\text{ }\mu\text{s}$. There are two MPTCP flows in the network, which use the same routing path as shown in Fig. 2. Among them, MPTCP 1 starts at 0 ms and continues transmitting data during the simulation time. MPTCP 2 starts at 4 ms and transmits 75 MB data in total.

Fig. 5 shows the real-time throughput of different algorithms under the scenario of Fig. 4(a). FCC utilizes a gradient

³In the simulation, the path's baseRTT is between $4\text{ }\mu\text{s}$ and $6\text{ }\mu\text{s}$. We set $\alpha_{total} = 1/0.005 = 200$ to achieve low queue length in the network, where the unit of RTT is ms. If the network condition changes, it should also change α_{total} to fit the network. For example, $\alpha_{total} = 2000$ is better for a DCN with a link bandwidth of 800 Gbps and base RTT of $0.1\text{ }\mu\text{s}$.

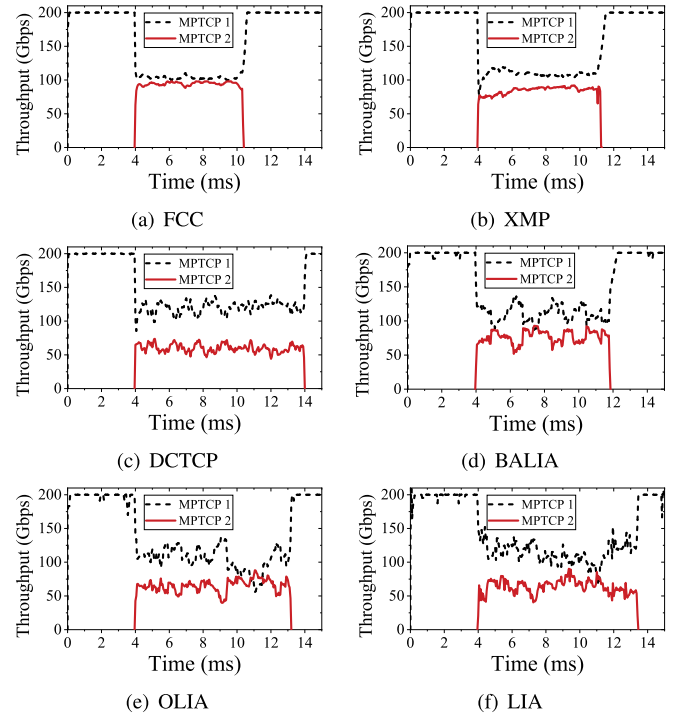


Fig. 5. Real time throughput of different congestion control algorithms under a non-optimal routing scenario.

TABLE II
THROUGHPUT COMPARISON OF DIFFERENT ALGORITHMS

Throughput (Gbps)	FCC	XMP	DCTCP	BALIA	OLIA	LIA
MPTCP 1	101.1	111.6	104.5	110.9	110.1	109.9
MPTCP 2	98.3	80.2	62.1	76.1	68.2	64.1
All	199.4	191.8	166.6	187.0	179.3	174.0
Utilization	99.7%	95.8%	83.3%	93.5%	89.6%	87%

projection method to adjust its rate according to the internal feedback information of the network, so it can quickly achieve the optimal load balancing effect, and thus maintains a stable transmission rate. Due to the strong fluctuation of packet loss feedback in the network, the real-time throughput of BALIA, OLIA, and LIA fluctuates greatly. Among them, BALIA performs better than OLIA and LIA, since it is designed for the network utility optimization. However, BALIA's throughput fluctuations are also high due to the greater impact of packet loss uncertainty. FCC, XMP, and DCTCP are based on ECN feedback, which have lower transmission fluctuations than BALIA, OLIA, and LIA. The FCC employs ECN to provide precise feedback within the network and dynamically adjusts congestion windows to optimize network utility, thereby maximizing the utilization of network resources. Both MPTCP flows achieve a throughput of 100 Gbps, resulting in near 100% overall bandwidth utilization.

Table II and Table III show the average throughput and RTT of individual MPTCP flows between 4 ms and 10 ms when both flows are concurrently transmitting data. As discussed earlier in our analysis, it was stated that the ideal solution has the capability to attain a maximum throughput of 100 Gbps for both of the MPTCP flows. According to the experimental results, it can be observed that FCC attains a performance level

TABLE III
RTT COMPARISON OF DIFFERENT ALGORITHMS

RTT (μ s)	FCC	XMP	DCTCP	BALIA	OLIA	LIA
MPTCP 1	5.07	5.94	6.23	10.54	9.69	9.26
MPTCP 2	5.25	6.81	6.84	12.71	12.52	13.57
Average	5.16	6.38	6.53	11.62	11.61	11.96

that is almost optimal, achieving a network bandwidth resource utilization of up to 99.7%. Furthermore, the rate allocation of the two MPTCP flows in the network remains well-balanced. Compared with other algorithms, DCTCP is a single-path congestion control algorithm that does not couple the multi-path transmission among different subflows, so the gap between the two MPTCP flows is the largest, and the network bandwidth resource utilization is the lowest. The throughput performance of XMP, BALIA, OLIA, and LIA is better than that of DCTCP, but the network bandwidth resources are still not fully utilized. In contrast to the aforementioned solutions, FCC ensures a fair distribution of resources between the two MPTCP flows and achieves a significantly high network bandwidth resource utilization of 99.7%. Moreover, ECN-based algorithms (DCTCP, XMP, FCC) can reduce the queue length at switches, which lead to much less RTT compared with loss-based algorithms (BALIA, OLIA, LIA). Out of all the solutions, FCC maintains the lowest average RTT.

Next, we present a performance analysis of a situation where four MPTCP flows and one single-path TCP flow contend at two bottleneck points. As illustrated in Figure 4(b), the network configuration includes four switching nodes and five pairs of source and destination nodes utilizing MPTCP and TCP. The MPTCP source servers establish connections to both switch A and switch C concurrently via distinct interfaces, while the destination servers connect to switch B and switch D. Every MPTCP source and destination pair creates an MPTCP connection with two subflows, each following a distinct path between the nodes of the MPTCP source and destination. Specifically, the TCP source connects to switch C, while the server connects to switch D. The bandwidth of each link is set to 100 Gbps, and the link delay is set to 1 μ s. During the transmission, MPTCP flows 1-4 begin transmitting at 0, 2.5, 5, and 7.5 ms, respectively, and end at 30, 27.5, 25, and 22.5 ms, respectively. The single-path TCP flow starts at 10 ms and ends at 20 ms. Fig. 6 shows the real-time throughput of different congestion control algorithms.

Out of various congestion control algorithms, LIA, OLIA, and BALIA modify the congestion window according to packet loss, resulting in significant fluctuations in throughput due to the stochastic nature of packet loss. The results shown in Figure 6(e) and Figure 6(f) demonstrate that the fluctuation in real-time throughput exceeds 50 Gbps. DCTCP, XMP, and FCC utilize ECN feedback and exhibit lower throughput variations, resulting in significantly reduced real-time throughput fluctuations compared to LIA, OLIA, and BALIA. As DCTCP is specifically intended for single-path transmission, it is incapable of maintaining fairness between TCP and MPTCP flows when they terminate simultaneously. Both AMP and FCC exhibit stability and fairness. However, FCC outperforms

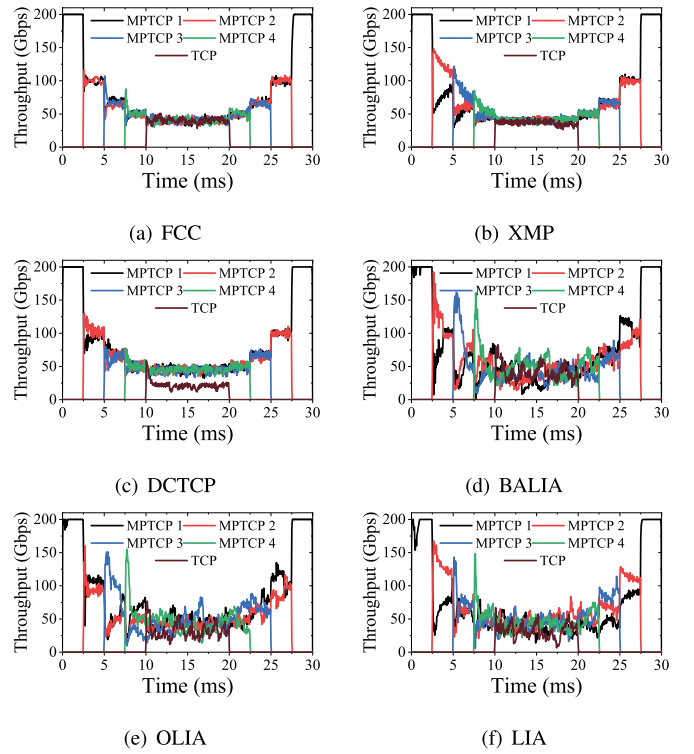


Fig. 6. The actual throughput of five flows contending for the link bandwidth in the scenario depicted in Fig. 4(b).

TABLE IV
AVERAGE THROUGHPUT BETWEEN 10 AND 20 MS

Throughput (Gbps)	FCC	XMP	DCTCP	BALIA	OLIA	LIA
MPTCP 1	40.26	40.99	45.65	37.18	42.46	39.45
MPTCP 2	39.76	40.93	44.83	38.30	39.91	46.52
MPTCP 3	40.34	40.79	43.35	39.91	41.75	41.95
MPTCP 4	39.71	41.54	44.29	41.63	41.24	39.85
TCP	40.19	35.72	21.85	41.16	34.11	32.20

AMP in terms of convergence time. This is due to the fact that XMP still uses a multiplicative decrease approach for the congestion window when receiving ECN-marked ACKs, whereas FCC uses a gradient method, resulting in improved convergence in both the congestion window and throughput.

Furthermore, between 10 and 20 ms, five flows transmit data concurrently. The bottleneck links have a total bandwidth of 200 Gbps, and each flow must receive 40 Gbps to ensure fairness. The average throughput of each flow is presented in Table IV. In terms of long-term average throughput, both FCC and BALIA exhibit the most favorable fairness by eventually apportioning 40 Gbps of bandwidth to each flow.

The RTT distribution of various algorithms is depicted in Fig. 7, showing that the FCC algorithm has the shortest RTT among all the algorithms. With a base RTT of 6 μ s, the LIA, OLIA, and BALIA algorithms exhibit higher RTTs, averaging around 14 μ s, which is twice the base RTT. On the other hand, DCTCP and XMP are able to decrease latency by utilizing ECN feedback, but they require a high ECN feedback threshold to prevent network underload, resulting in an average RTT of approximately 9 μ s. In comparison to

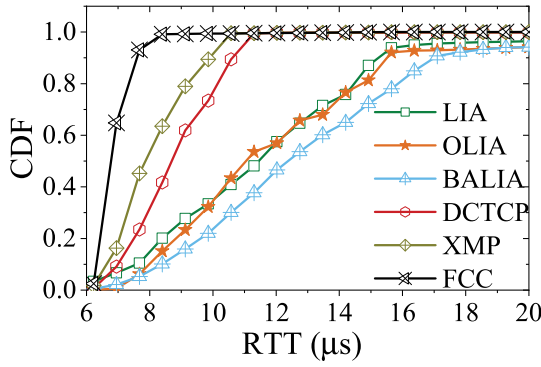


Fig. 7. Distribution of real-time RTT.

TABLE V
AVERAGE QUEUE OCCUPANCY RATIO ON SWITCH A

Algorithm	FCC	XMP	DCTCP	BALIA	OLIA	LIA
Queue occupancy (%)	6.7	16.05	13.09	49.63	43.14	42.75

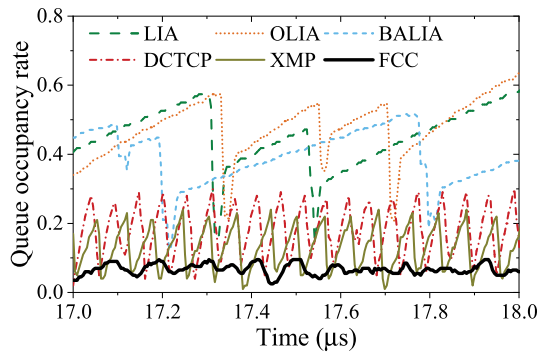


Fig. 8. Real-time queue occupancy rate.

these algorithms, FCC offers even lower RTT and guarantees throughput performance.

Table V and Fig. 8 show the average and real-time queue occupancy at switch A, respectively. LIA, OLIA, and BALIA generate lengthy queues at the switch, resulting in significant fluctuations and subsequently leading to high RTT and RTT jitter. In contrast, DCTCP and XMP exhibit better performance than LIA and BALIA, reducing the average queue occupancy by over 50%. However, XMP still presents substantial queue occupancy fluctuation. FCC outperforms all other algorithms by reducing the average queue occupancy by 86.5% compared to BALIA and 62.3% compared to XMP. Additionally, FCC demonstrates the least fluctuation in queue occupancy rate at the switch, as it does not halve the congestion window.

B. Congestion Control and Routing in Large Scale Networks

We then evaluate the performance for large-scale DCN scenarios using a 3-tier FatTree topology, where $K = 10$, including 125 switches and 500 servers. The bandwidth of each link is set to 1 Gbps, and the link delay is set randomly between 0.02 ms and 0.04 ms. The algorithms AMP, BALIA, and LIA are utilized as benchmarking techniques for congestion control in multi-path scenarios. AMP is similar to XMP, but adds subflow management to deal with the incast problem

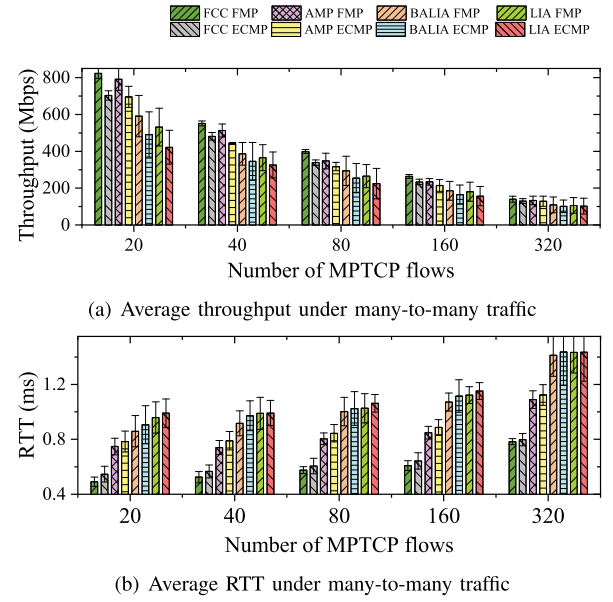


Fig. 9. Many-to-many traffic under a FatTree topology.

in DCNs. Therefore, we choose AMP for the comparison in a large-scale DCN. The parameters used in FCC are set to $high_thresh = 0.8$, $\alpha_{total} = 1$, $w_{min} = 10$, and the maximum queue length of each port of the switches is set to 100 packets.

The FCC and FMP algorithms of FMPTCP can be used separately and work with other congestion control and routing algorithms. Hence, we conduct simulations for various combinations of multi-path routing and congestion control algorithms. FMP and ECMP are utilized as the routing algorithms, while LIA, BALIA, AMP, and FCC are used as the congestion control algorithms. We evaluate the performance in scenarios involving many-to-one, many-to-many, and random traffic, and analyze the impact of varying numbers of subflows. Eight different combinations are simulated, and the performance of each algorithm is presented based on the average throughput and average RTT.

1) *Many-to-Many Traffic*: Each MPTCP flow chooses one of the first 20 servers as the source and one of the last 20 servers as the destination to create an MPTCP connection with 4 subflows. Each MPTCP flow transmits 50 MB of data. The average throughput and RTT performance of various algorithms are depicted in Fig. 9, as the number of MPTCP flows changes. In this evaluation, each MPTCP flow transmits 50 MB of data, and the interval between two MPTCP connection establishments is set to 5 ms.

As the number of MPTCP flows increases, the average throughput gradually decreases at a slower rate than in many-to-one traffic. Fig. 9(a) shows that LIA with ECMP typically results in the lowest throughput, while BALIA and AMP perform better but still fail to fully utilize network resources. Among the congestion control algorithms, FCC maintains a higher average throughput than the others by contributing to load balancing. This is because FCC is better at utilizing network resources. Furthermore, FMP further improves the throughput compared to ECMP when used with different congestion control algorithms, as it takes into account feedback

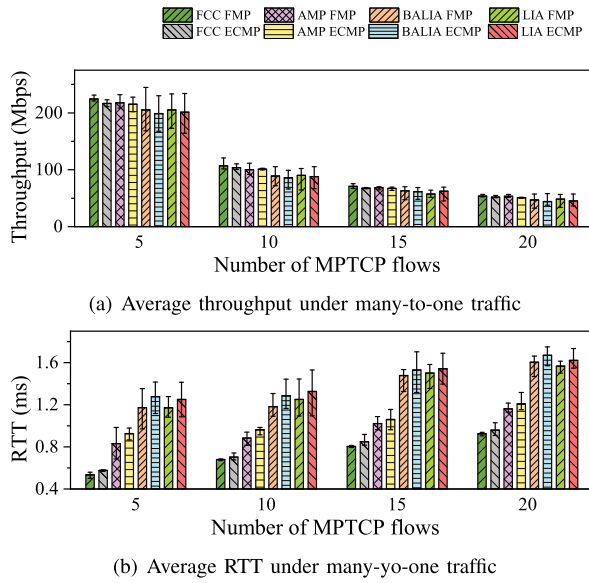


Fig. 10. Many-to-one traffic under a FatTree topology.

information regarding congestion and delay in the network. Similarly, as demonstrated in Fig. 9(b), FCC with FMP results in the lowest average RTT. Compared to other algorithms that continuously increase the congestion window, leading to a large queue and collective packet loss in the network, FCC maintains a stable transmission rate and reduces the queue length in the network, which results in a low RTT. In addition, FMP selects less congested and delayed paths, further reducing the RTT. FCC with FMP reduces the RTT by 50% compared to LIA ECMP/FMP and BALIA ECMP/FMP, and reduces the RTT by 20% compared to AMP ECMP/FMP. Moreover, when there are 320 flows, the network is highly congested. It is shown that the RTT of loss-based congestion control algorithms (BALIA, LIA) increases a lot. AMP's RTT is smaller than BALIA, LIA but is still increased. FCC's RTT increases slightly. In this scenario, the difference between routing algorithms is little.

2) *Many-to-One Traffic*: When dealing with many-to-one traffic, the last server is selected as the destination. Each time a new flow begins, a random server (excluding the destination server) is chosen as the source, and an MPTCP connection is established with 4 subflows between the source and destination servers. Random data sizes between 5 and 200 MB are transmitted through each MPTCP flow. Once a flow is completed, a new source server is chosen to start another flow after 5 ms. Figure 10 displays the average throughput and RTT performance of the network, with the maximum number of flows transmitting data simultaneously. In the case of many-to-one traffic, the performance of a single MPTCP flow deteriorates as the total number of flows increases because all flows will eventually converge on the switch accessed by the source server. FCC with FMP, however, attains the highest throughput and the lowest RTT in this scenario.

The performance of throughput is presented in Fig. 10(a), which demonstrates that FCC maintains a higher average throughput than other algorithms. As all the MPTCP flows converge at the last edge switch, FMP does not exhibit its

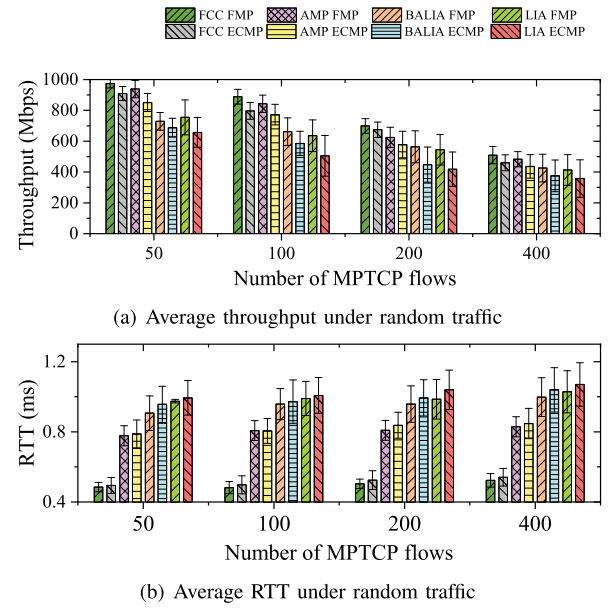


Fig. 11. Random traffic under a FatTree topology.

superiority in such cases. Fig. 10(b) shows the performance of average RTT, which increases rapidly as the number of flows increases. LIA and BALIA continuously send packets, filling the switch's queue until packet loss occurs, leading to increased delay. In contrast, AMP utilizes ECN feedback at a fixed queue length on the switch, resulting in lower RTT than LIA and BALIA. When only five flows are present, FCC achieves a much lower RTT than other algorithms. With an increase in the number of flows, the average RTT of FCC also increases rapidly, yet it remains lower than others. Additionally, FMP helps reduce RTT in this situation. Compared to ECMP, FMP reduces RTT for all congestion control algorithms.

3) *Random Traffic*: In order to emulate a random traffic scenario, a pair of servers are randomly selected to establish an MPTCP connection with 4 subflows. Each subflow transmits 50 MB of data, and the time interval between two MPTCP connections being established is set to 5 ms. Figure 11 displays the average throughput and RTT performance of various algorithms with varying numbers of MPTCP flows. The number of MPTCP flows is indicative of the network traffic density.

When the number of MPTCP flows in the network is less than 100, there is sufficient bandwidth for each subflow to achieve the maximum throughput of 1 Gbps. Among the various algorithms, FCC with FMP achieves the highest throughput, almost reaching 1 Gbps, and fully utilizes the available bandwidth. Although FCC with ECMP has a lower throughput than FCC with FMP, it still outperforms AMP, BALIA, and LIA. As the number of MPTCP flows increases, the average throughput gradually decreases, but FCC with FMP still maintains a higher throughput than the other algorithms. Figure 11(b) shows that FCC with FMP also has the lowest average RTT.

4) *Changing the Number of Subflows*: Multiple subflows are utilized by MPTCP to enhance the utilization of bandwidth in DCNs. A high number of subflows can distribute

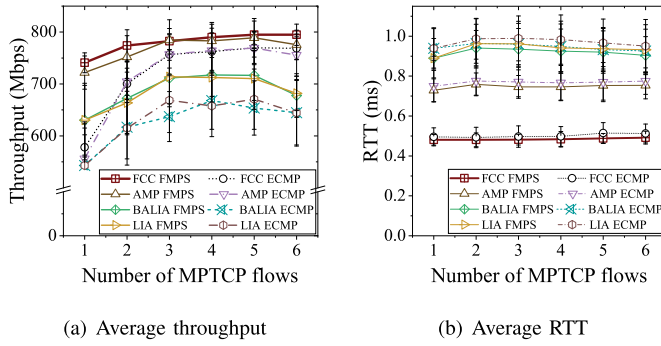


Fig. 12. Performance evaluation with changing subflows.

traffic across all links in the network and result in increased resource utilization. Therefore, we also evaluate performance by simulating various numbers of subflows. For this scenario, a random traffic pattern is utilized with a total of 150 MPTCP flows, while the number of subflows for each MPTCP flow varies between 1 and 6.

Both FCC and AMP exhibit similar average throughput performance, regardless of whether FMP or ECMP is used, and outperform LIA and BALIA. FMP further enhances average throughput compared to ECMP. As the number of subflows increases, the average throughput of FCC+ECMP and AMP+ECMP steadily increases, with a 41.5% increase in throughput observed when using 6 subflows compared to only 1 subflow. The throughput of LIA+ECMP and BALIA+ECMP initially increases but then decreases. This is because they do not distribute traffic well across subflows, resulting in reduced overall network throughput. FMP selects paths with lower congestion for flows, allowing it to achieve higher throughput with fewer subflows than ECMP. Figure 12(a) displays the average RTT, which slightly increases with the number of subflows for different algorithms, while FCC with FMP consistently maintains the lowest RTT. Compared to other algorithms, FMPTCP (FCC+FMP) enhances throughput by 12.4-35.9% and reduces transmission latency by 32.5-51.2% in DCNs.

C. Summary

In conclusion, the experiments revealed several findings and suggestions. Firstly, the effective utilization of network resources is a complex issue that requires a joint solution involving both end-to-end multi-path transmission and in-network routing. While each aspect can improve performance to a certain extent, combining both is necessary to optimize overall performance. Achieving joint optimization in the Internet is almost impossible due to the complexity and uncontrollability of the devices. However, in a DCN where all nodes are controllable and modifiable, such joint optimization can be achieved to improve the performance of the entire network. This kind of joint optimization is especially important for DCNs, as the application performance of distributed computing and distributed storage heavily relies on the efficient utilization of network resources. Secondly, as the scale of DCNs increases, the significance of routing in joint optimization will also increase. End-to-end multi-path

transmission can only migrate traffic on established subflows on some paths, leaving the remaining paths underutilized. With the exponential increase in network scale, high-efficient, high-precision, and fast-reactive routing will become a key factor affecting DCN transmission performance. Lastly, the experiment on the different number of subflows revealed that increasing the number of subflows can improve the resource utilization efficiency of the entire network. However, blindly increasing the number of subflow does not necessarily lead to performance improvement, beyond which the performance improvement becomes minimal, but the overhead of managing subflows still increases. Therefore, determining the optimal number of subflows is a complicated issue that involves a trade-off between performance and overhead and also depends on network scale and traffic characteristics. This issue deserves the attention of researchers and should be considered in further design. While commercial data centers typically have tens of thousands of servers, it is worth noting that existing experiments have provided sufficient insights into the transmission issues within DCNs. Furthermore, in larger-scale topologies, it is indeed expected that a higher number of subflows would be required to achieve optimal performance.

VII. CONCLUSION

We presented a theoretical analysis of the MPTCP resource pooling problem in DCNs, focusing on rate allocation and multi-path routing sub-problems. Based on the analysis results, we proposed FMPTCP, a feedback-based MPTCP solution that incorporates FCC and FMP. FCC utilizes ECN for precise feedback on in-network congestion to achieve optimal rate allocation and low latency under a fixed routing topology. FMP proposes a hop-by-hop feedback and routing approach to further improve resource utilization in DCNs. Simulation results demonstrated that FMPTCP significantly improves transmission performance in DCNs, outperforming previous algorithms by increasing throughput by 12.4-35.9% and reducing transmission latency by 32.5-51.2%. These findings suggest that FMPTCP represents a promising solution for the resource pooling problem in DCNs, improving overall network performance and facilitating the efficient utilization of network resources.

ACKNOWLEDGMENT

The authors would like to express their sincere gratitude to the anonymous referees for their valuable suggestions that have resulted in an improved version of the original manuscript.

APPENDIX

A. Proof of Proposition 1

According to KKT Conditions [48], the solution simultaneously satisfies:

$$\frac{\rho_{\min}^k}{\ell_j} U'_k(y_k) - q_j \leq 0, \quad (21)$$

$$x_{jk} \geq 0, \quad (22)$$

$$\left(\frac{\rho_{\min}^k}{\ell_j} U'_k(y_k) - q_j \right) x_{jk} = 0. \quad (23)$$

TABLE VI

THROUGHPUT COMPARISON WITH DIFFERENT FCC PARAMETERS

Throughput (Gbps)	$\alpha_{total} = 200$ $hh = 80 \text{ pkts}$	$\alpha_{total} = 400$ $hh = 80 \text{ pkts}$	$\alpha_{total} = 200$ $hh = 200 \text{ pkts}$
MPTCP 1	101.1	103.9	104.1
MPTCP 2	98.3	94.2	94.8
All	199.4	198.1	198.9
Utilization	99.7%	99.1%	99.5%

TABLE VII

RTT COMPARISON WITH DIFFERENT FCC PARAMETERS

RTT (μ s)	$\alpha_{total} = 200$ $hh = 80 \text{ pkts}$	$\alpha_{total} = 400$ $hh = 80 \text{ pkts}$	$\alpha_{total} = 400$ $hh = 200 \text{ pkts}$
MPTCP 1	5.07	5.46	6.11
MPTCP 2	5.25	5.67	6.48
Average	5.16	5.55	6.39

Using Eq. (6) to substitute Eq. (21), Eq. (22), and Eq. (23). Then, we have $\frac{\ell_{\min}^k}{\ell_j} U'_k(y_k) - q_j \leq \frac{\ell_{\min}^k}{\ell_j} (U'_k(y_k) - \min_{j \in \mathcal{J}_k} \frac{\ell_j}{\ell_{\min}^k} q_j) = 0$. Therefore, Eq. (21) holds. When $x_{jk} = 0$, $\frac{\ell_{\min}^k}{\ell_j} U'_k(y_k) - q_j < 0$, and when $x_{jk} > 0$, $\frac{\ell_{\min}^k}{\ell_j} U'_k(y_k) - q_j = 0$. Therefore, Eq. (22) and Eq. (23) hold. Then, the solution of $D_k(\lambda)$ satisfies Eq. (6).

B. Proof of Proposition 2

We show a simple proof of Proposition 2. Given a feasible solution $\mathbf{x}$ of Eq. (10), $\mathbf{x}$ is also a feasible solution of Eq. (2) with $\mathbf{B} = \mathbf{F}(\mathbf{x}) * \mathbf{G}$. Therefore, $U(\mathbf{F}(\mathbf{x})) = U(\mathbf{x}^*(\mathbf{F}(\mathbf{x}))) \geq U(\mathbf{x})$. Let $\mathbf{F}_{opt}$ be the optimal solution of Eq. (9), $U(\mathbf{F}_{opt})$ be the optimal utility of Eq. (9), and $U(\mathbf{x}_{opt})$ be the optimal utility of Eq. (10), we have $U(\mathbf{x}_{opt}) \leq U(\mathbf{F}(\mathbf{x}_{opt})) \leq U(\mathbf{F}_{opt})$. Given a feasible solution $\mathbf{F}$ of (9), $\mathbf{x}^*(\mathbf{F})$ is also a feasible solution of Eq. (10) and $U(\mathbf{x}^*(\mathbf{F}_{opt})) = U(\mathbf{F}_{opt})$. Therefore, $U(\mathbf{F}_{opt}) = U(\mathbf{x}^*(\mathbf{F}_{opt})) \leq U(\mathbf{x}_{opt})$. Then, we have $U(\mathbf{x}_{opt}) = U(\mathbf{F}_{opt})$ and $\mathbf{F}_{opt} = \mathbf{F}(\mathbf{x}_{opt})$.

C. Comparison of Different FCC Parameter Settings

The transmission performance of FCC is influenced by its parameter settings. To demonstrate this, we present a comparison of throughput and RTT with different parameter values in Figure 4(a), where we vary the values of α_{total} and $high_thresh$ (hh for brevity). The maximum queue length is set to 300 packets, and all other parameters are kept the same as in Section VI. We observe that the MPTCP throughput remains relatively constant across the different parameter settings. However, the RTT exhibits significant variation, with larger values of α_{total} and $high_thresh$ leading to increased RTT.

D. Armijo Rule in Gradient Projection

Compared with the standard gradient projection, some efficient solution algorithms, such as gradient projection based on Armijo rule, provide a faster convergence time with high computational complexity. Figure 13 shows a comparison between FCC using the standard gradient projection and FCC using the Armijo rule. The results show that FCC with Armijo

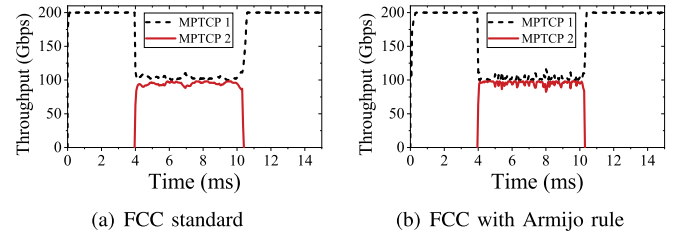


Fig. 13. Comparison of throughput with different gradient projection methods.

rule converges faster than the standard FCC, although the standard FCC already has good performance. Since FCC employs a per-ACK window adjustment, it uses the standard gradient projection to balance the convergence performance and computational complexity.

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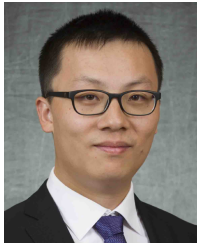


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